THE LIMIT OF THE STRUCTURE SHEAF ALONG THE FROBENIUS IS A SHEAF FOR THE H TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVL (sources/stacks-flat/pinned-original-source.tex line 11931).

Allowed standard prerequisites (editorial).

- The four conditions characterising h sheaves. Same chapter, Lemma labelled lemma-refine-check-h, cited at proof lines 11941-11943. Provides the checklist verified in the proof.
- The fppf sheaf property of the structure sheaf. Descent, Lemma labelled descent-lemma-sheaf-condition-holds, cited at proof lines 11945-11947. Settles condition (2) together with commutation of the limit with equalizers.
- The one generator example and the nilpotence of fA cap J. Same chapter, Example labelled example-one-generator, cited at proof lines 11953-11955. Supplies the algebra setting of condition (3).
- The two generators example and the funny blowup. Same chapter, Example labelled example-two-generators, invoked at proof lines 11969-11971, and Lemma labelled lemma-funny-blow-up, cited at proof lines 11969-11973. Settles condition (4).

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVI: sources/stacks-flat/pinned-original-source.tex, source lines 11501–11501. Same-chapter pointer into the author environment lemma at source line 11501; invoked at proof line 11943 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03DT: sources/stacks-chow/tags.txt, source lines 4432–4432. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 11947 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0EVG: sources/stacks-flat/pinned-original-source.tex, source lines 11474–11474. Same-chapter pointer into the author environment example at source line 11474; invoked at proof line 11953 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0EVH: sources/stacks-flat/pinned-original-source.tex, source lines 11488–11488. Same-chapter pointer into the author environment example at source line 11488; invoked at proof line 11972 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVJ: sources/stacks-flat/pinned-original-source.tex, source lines 11782–11782. Same-chapter pointer into the author environment lemma at source line 11782; invoked at proof line 11973 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let p be a prime number. Let S be a scheme over $\mathbf{F}_p$. Let $(Sch/S)_h$ be a site as in Definition 0EU1. The rule*

$$\mathcal{F}(X) = \lim_F \Gamma(X, \mathcal{O}_X)$$

defines a sheaf on $(Sch/S)_h$.

Proof. To prove $\mathcal{F}$ is a sheaf, let's check conditions (1), (2), (3), and (4) of Lemma 0EVI. Condition (1) holds because limits of sheaves are sheaves and $\mathcal{O}$ is a Zariski sheaf. Condition (2) holds because $\mathcal{O}$ is an fppf sheaf (Descent, Lemma 03DT) and if A is the equalizer of two maps $B \to C$ of $\mathbf{F}_p$-algebras, then $\lim_F A$ is the equalizer of the two maps $\lim_F B \to \lim_F C$.

We check condition (3). Let A, f, J be as in Example 0EVG. We have to show that

$$\begin{aligned} \lim_F A &\to \lim_F A/J \times_{\lim_F A/fA+J} \lim_F A/fA \\ &= \lim_F (A/J \times_{A/fA+J} A/fA) \\ &= \lim_F A/(fA \cap J) \end{aligned}$$

is bijective. Since J is annihilated by a power of f we see that $\mathfrak{a} = fA \cap J$ is a nilpotent ideal, i.e., there exists an n such that $\mathfrak{a}^n = 0$. It is straightforward to verify that in this case $\lim_F A \to \lim_F A/\mathfrak{a}$ is bijective.

We check condition (4). Let X, X', Z, E be as in Example 0EVH. By Lemma 0EVJ and the same argument as above we see that

$$\mathcal{F}(X) = \lim_F \Gamma(X, \mathcal{O}_X) \longrightarrow \lim_F \Gamma(X', \mathcal{O}_{X'}) = \mathcal{F}(X')$$

is bijective. Since $E = \mathbf{P}^1_Z$ in this case we also see that $\mathcal{F}(Z) \to \mathcal{F}(E)$ is bijective. Thus the conclusion holds in this case as well. □